

DATA STRUCTURE LAB

(DCO – 414)

Practical No. 3: Write a program in C to add n numbers using pointers.

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-----ALGORITHM-----

- STEP 1:** Start
- STEP 2:** Read the value of n
- STEP 3:** Declare pointer int *p, integer i, and sum = 0
- STEP 4:** Allocate memory dynamically using malloc() for n integers
- STEP 5:** Check if memory allocation is successful (p != NULL)
- STEP 6:** If memory is not allocated, display error message and Stop
- STEP 7:** Input n numbers using pointer (p+i)
- STEP 8:** Add all numbers using $sum = sum + *(p+i)$
- STEP 9:** Display the final value of sum
- STEP 10:** Free the allocated memory using free(p)
- STEP 11:** Stop

-----CODE-----

```
#include<stdio.h>
#include<stdlib.h>
int main()
{
    int n,*arr,i,sum=0;
    printf("ENTER THE SIZE OF THE ARRAY:");
    scanf("%d",&n);
    arr=(int *)calloc(n,sizeof(int));
    if(arr==NULL)
    {
        printf("MEMORY ALLOCATION FAILED");
        return 0;
    }
    printf("ENTER THE VALUES:\n");
    for (i=0;i<n;i++)
    {
        printf("VALUE AT %d is:",i+1);
        scanf("%d",&arr[i]);
    }
    printf("ENTERED ARRAY IS:\n");
    for (i=0;i<n;i++)
    {
        printf("%d\t",*(arr+i));
    }
    for (i=0;i<n;i++)
    {
        sum+=*(arr+i);
    }
    printf("\nSUM OF %d NUMBERS IS = %d",n,sum);
    return 0;
}
```

-----OUTPUT-----

ENTER THE SIZE OF THE ARRAY:5

ENTER THE VALUES:

VALUE AT 1 is:11

VALUE AT 2 is:23

VALUE AT 3 is:15

VALUE AT 4 is:21

VALUE AT 5 is:65

ENTERED ARRAY IS:

11 23 15 21 65

SUM OF 5 NUMBERS IS = 135